

PROJECT REPORT
ON
“CYBERSECURITY RISK ASSESSMENT FRAMEWORK FOR SMEs”

Internship Project Report Submitted for
Cyber Security Internship Program

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Abstract

This project presents a cybersecurity risk assessment framework designed for small and medium enterprises (SMEs). The project identifies common cybersecurity threats such as phishing, ransomware, malware, weak passwords, and insider threats. A risk assessment matrix was developed to evaluate threats based on likelihood and impact. The project also applies the NIST Cybersecurity Framework to propose effective mitigation strategies including firewall protection, antivirus software, multi-factor authentication, secure network configurations, and employee cybersecurity awareness training. Case studies were conducted on different SMEs to analyze cybersecurity vulnerabilities and recommend appropriate security measures.

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1. Introduction

Cybersecurity has become one of the most important concerns for modern businesses due to the increasing number of cyberattacks and data breaches. Small and medium enterprises (SMEs) are highly vulnerable to cybersecurity threats because they often lack strong security infrastructure, technical expertise, and cybersecurity awareness. Cyberattacks such as phishing, ransomware, malware, and unauthorized access can cause financial losses, operational disruption, and reputational damage. Therefore, organizations must identify cybersecurity risks and implement effective security controls to protect sensitive information and business systems.

2. Objectives

- To identify common cybersecurity threats affecting SMEs.
- To develop a cybersecurity risk assessment framework.
- To evaluate cybersecurity risks using a risk matrix model.
- To recommend mitigation strategies for improving cybersecurity.
- To analyze cybersecurity challenges through case studies.
- To promote cybersecurity awareness and best practices.

3. NIST Cybersecurity Framework

The National Institute of Standards and Technology (NIST) developed the NIST Cybersecurity Framework to help organizations identify, manage, and reduce cybersecurity risks. The framework provides a structured approach that helps businesses strengthen their cybersecurity posture and respond effectively to cyber threats.

The NIST Cybersecurity Framework is widely used by organizations because it provides clear guidelines for protecting systems, networks, and sensitive business information. It consists of five core functions that work together to improve cybersecurity management and risk handling within organizations.

For small and medium enterprises (SMEs), the framework helps in identifying vulnerabilities, implementing protective security controls, detecting cyber incidents, responding to attacks, and recovering from security breaches efficiently.

Table 1: Core Functions of the NIST Cybersecurity Framework

Function	Purpose
Identify	Identify cybersecurity risks
Protect	Implement security controls
Detect	Detect cyber threats
Respond	Respond to incidents
Recover	Restore operations after attack

In this project, the NIST framework was used as the foundation for evaluating cybersecurity risks and proposing mitigation strategies for SMEs.

4. Risk Assessment Matrix

The risk assessment matrix was designed to analyze the most common cybersecurity threats affecting small and medium enterprises (SMEs). Each threat was evaluated based on two important factors: likelihood and impact. Likelihood represents the probability of a cyberattack occurring, while impact represents the level of damage it may cause to business operations, financial records, or sensitive information. By calculating the risk score, organizations can identify critical threats and prioritize cybersecurity measures accordingly.

The risk score was calculated using the following formula:

$$\text{Risk Score} = \text{Likelihood} \times \text{Impact}$$

The calculated risk score helps organizations prioritize cybersecurity threats based on their severity and potential business impact.

Table 2: Cybersecurity Risk Assessment Matrix

Threat	Likelihood	Impact	Risk Score	Severity
Phishing	5	4	20	Critical
Ransomware	4	5	20	Critical
Weak Passwords	5	3	15	High
Malware	4	4	16	High
Insider Threat	3	5	15	High
Unsecured Wi-Fi	3	4	12	Medium

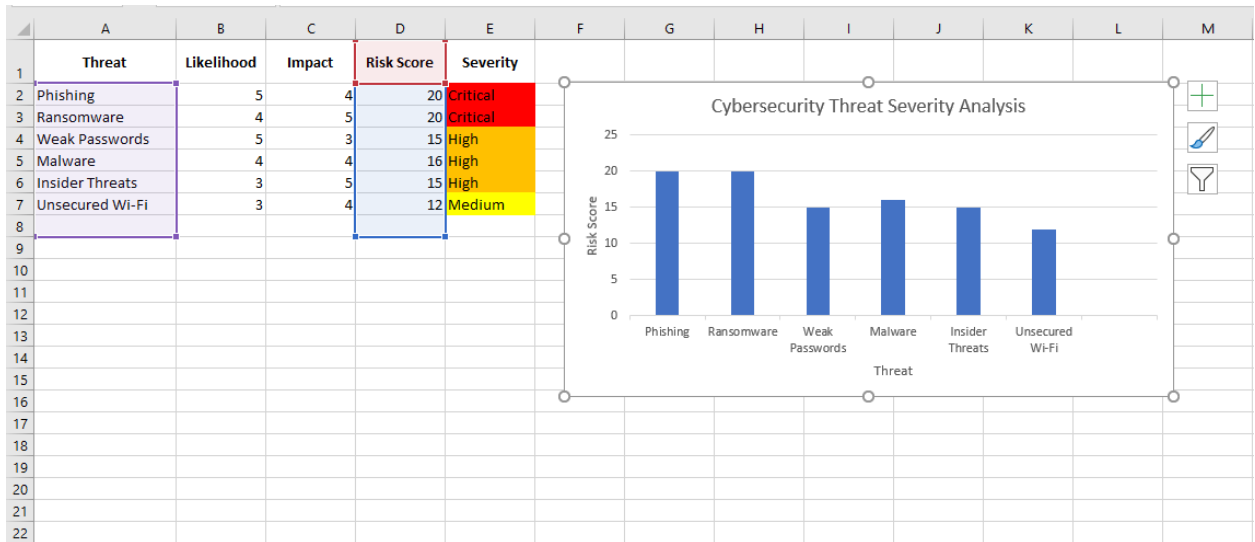


Figure 1: Cybersecurity Threat Severity Analysis

The analysis shows that phishing attacks and ransomware represent the highest cybersecurity risks for SMEs due to their high likelihood and severe impact on business operations and sensitive data.

5. Case Study

The following case studies demonstrate how cybersecurity threats can affect different types of SMEs and how appropriate mitigation strategies can reduce organizational security risks.

5.1. Case Study: Medical Shop

The medical shop stores customer billing records and medicine inventory data on a shared computer system used by multiple employees.

Problems:

The shop uses weak passwords that are easy to guess. Multiple employees use the same system without proper login security. No antivirus software is installed on the computer system.

Risks:

Weak security practices can allow attackers to steal customer data or infect the system with ransomware. Since the computer is shared, unauthorized access becomes easier.

Mitigation:

To improve cybersecurity, the shop should install antivirus software, enable firewall protection, use strong passwords, and implement multi-factor authentication (MFA). Employees should also receive basic cybersecurity awareness training.

Table 3: Cybersecurity Risk Analysis of Medical Shop

Category	Details
Business Type	Medical Shop
Main Threat	Ransomware

Category	Details
Risk Level	High
Recommended Solution	MFA and Antivirus

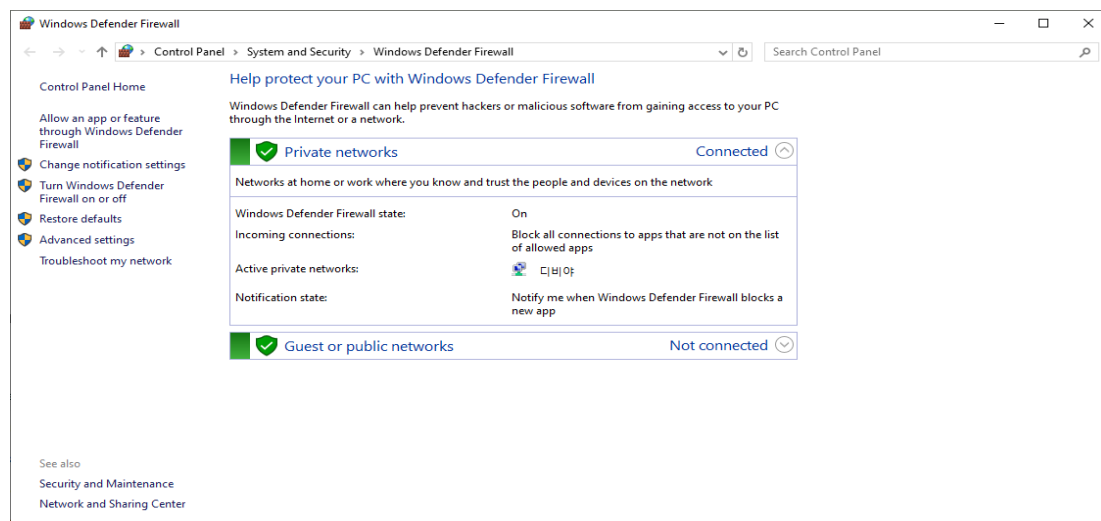


Figure 2: Windows Defender Firewall Protection

5.2. Case Study: Clothing Store

The clothing store uses online payment systems and provides WiFi access for business operations.

Problems:

The store uses an unsecured WiFi network without proper monitoring tools or firewall configuration.

Risks:

Hackers connected to the same network may intercept payment information or access store systems illegally.

Mitigation:

The business should secure the WiFi router using WPA2/WPA3 encryption, change default passwords, install firewall protection, and use VPN services for secure communication.

Table 4: Cybersecurity Risk Analysis of Clothing Store

Category	Details
Business Type	Clothing Store
Main Threat	Payment Theft
Risk Level	Medium
Recommended Solution	Secure WiFi and VPN

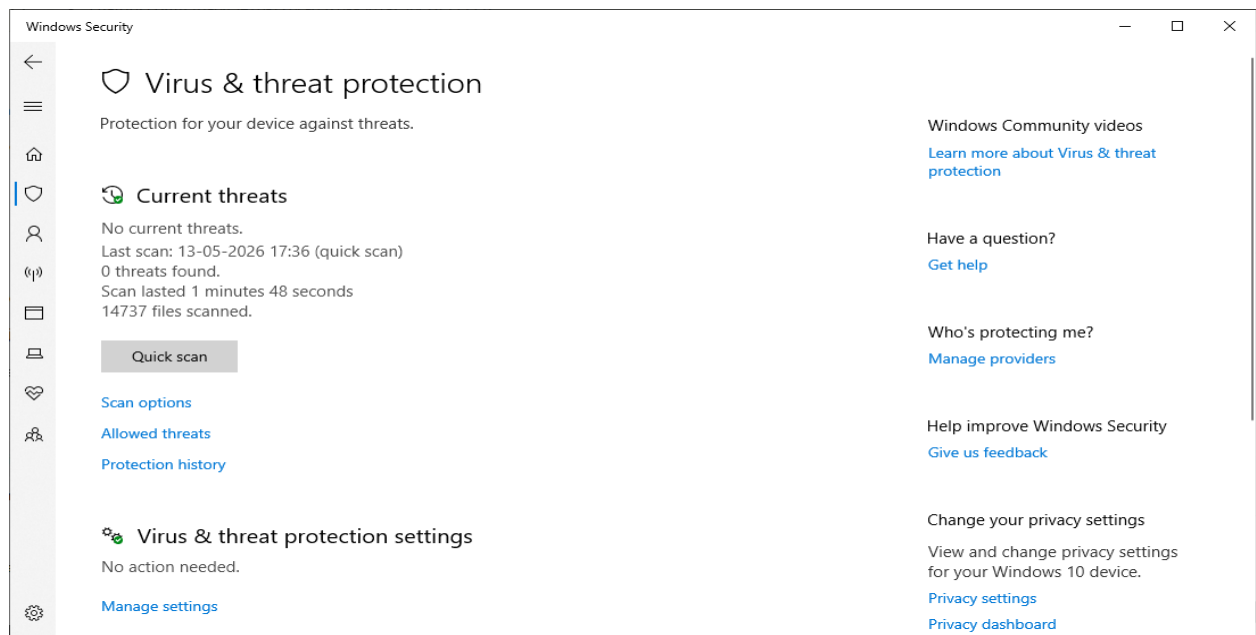


Figure 3: Windows Virus and Threat Protection Dashboard

5.3. Case Study: Coaching Institute

The coaching institute stores student records, fees data, and study material on shared systems and cloud platforms.

Problems:

Multiple staff members use shared login credentials to access administrative systems and cloud storage.

Risks:

Shared access increases the chances of data leakage, accidental file deletion, and unauthorized modification of student information.

Mitigation:

The institute should implement role-based access control, use separate login credentials for employees, enable strong password policies, and conduct cybersecurity awareness sessions.

Table 5: Cybersecurity Risk Analysis of Coaching Institute

Category	Details
Business Type	Coaching Institute
Main Threat	Data Breach
Risk Level	High
Recommended Solution	Access Control

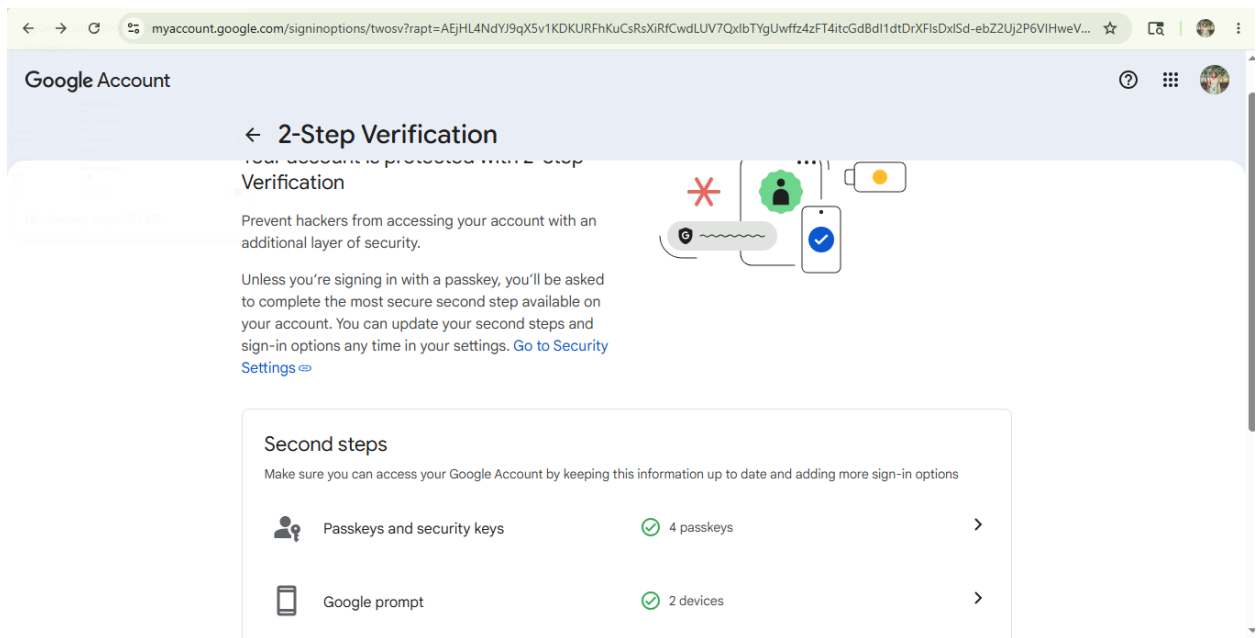


Figure 4: Google 2-Step Verification Security Setting

Table 6: Comparative Analysis of Cybersecurity Risks in SMEs

Business	Main Risk	Severity	Mitigation
Medical Shop	Ransomware	High	MFA + Antivirus
Clothing Store	Payment Theft	Medium	VPN + Secure WiFi
Coaching Institute	Data Breach	High	Access Control

These case studies demonstrate how cybersecurity threats can significantly impact small and medium enterprises (SMEs) and highlight the importance of proactive security measures.

6. Best Practices for SME Cybersecurity

- Use strong and unique passwords.
- Enable multi-factor authentication (MFA).
- Regularly update operating systems and software.
- Install antivirus and firewall protection.
- Conduct employee cybersecurity awareness training.
- Secure Wi-Fi networks using WPA2/WPA3 encryption.
- Regularly back up important business data.
- Restrict unauthorized system access.

7. Conclusion

Cybersecurity threats are rapidly increasing among small and medium enterprises (SMEs) due to limited security awareness and inadequate cybersecurity infrastructure. This project developed a cybersecurity risk assessment framework based on the NIST Cybersecurity Framework to identify common cybersecurity threats, evaluate risks, and recommend effective mitigation strategies. The findings demonstrate that implementing proactive security measures such as firewalls, antivirus software, multi-factor authentication, secure network configurations, and employee awareness training can significantly improve the cybersecurity posture of SMEs and reduce potential cyber risks.

8. References

1. National Institute of Standards and Technology (NIST) Cybersecurity Framework
2. Microsoft Security Documentation
3. Google Account Security Documentation
4. Cybersecurity and Infrastructure Security Agency (CISA)
5. Research articles on SME cybersecurity risks and mitigation strategies